

Enter password: ****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 9

Server version: 8.0.45 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> use rutuja;
```

Database changed

```
mysql> CREATE TABLE areas (
```

```
    -> radius INT,
```

```
    -> area FLOAT
```

```
    -> );
```

ERROR 1050 (42S01): Table 'areas' already exists

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE calculate_area()
```

```
    -> BEGIN
```

```
    -> DECLARE r INT;
```

```
    -> DECLARE a FLOAT;
```

```
    ->
```

```
    -> SET r = 5;
```

```

->
-> WHILE r <= 9 DO
->
->     SET a = 3.14 * r * r;
->
->     INSERT INTO areas(radius, area)
->     VALUES(r, a);
->
->     SET r = r + 1;
->
-> END WHILE;
->
-> END //

```

Query OK, 0 rows affected (0.02 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL calculate_area();

Query OK, 1 row affected (0.07 sec)

mysql> SELECT * FROM areas;

```
+-----+-----+
```

```
| radius | area |
```

```
+-----+-----+
```

```
| 5 | 78 |
```

```
| 6 | 113 |
```

```
| 7 | 154 |
```

```
| 8 | 201 |
```

| 9 | 254 |

+-----+-----+

5 rows in set (0.02 sec)

mysql> -- STEP 3 : Create Old RollCall Table

mysql>

mysql> CREATE TABLE O_RollCall (

-> roll_no INT PRIMARY KEY,

-> student_name VARCHAR(50)

->);

Query OK, 0 rows affected (0.08 sec)

mysql> -- STEP 4 : Create New RollCall Table

mysql>

mysql> CREATE TABLE N_RollCall (

-> roll_no INT PRIMARY KEY,

-> student_name VARCHAR(50)

->);

Query OK, 0 rows affected (0.17 sec)

mysql> INSERT INTO O_RollCall VALUES

-> (1,'Rutuja'),

-> (2,'Amit'),

-> (3,'Sneha');

Query OK, 3 rows affected (0.03 sec)

Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO N_RollCall VALUES

```
-> (2,'Amit'),  
-> (3,'Sneha'),  
-> (4,'Rahul'),  
-> (5,'Pooja');
```

Query OK, 4 rows affected (0.13 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE merge_rollcall()
```

```
-> BEGIN  
  
->  
  
-> DECLARE done INT DEFAULT 0;  
  
->  
  
-> DECLARE r_no INT;  
-> DECLARE s_name VARCHAR(50);  
  
->  
  
-> DECLARE cur CURSOR FOR  
-> SELECT roll_no, student_name  
-> FROM N_RollCall;  
  
->  
  
-> DECLARE CONTINUE HANDLER FOR NOT FOUND  
-> SET done = 1;  
  
->  
  
-> OPEN cur;  
  
->  
  
-> read_loop: LOOP  
  
->
```

```

->    FETCH cur INTO r_no, s_name;
->
->    IF done = 1 THEN
->        LEAVE read_loop;
->    END IF;
->
->    IF NOT EXISTS (
->        SELECT *
->        FROM O_RollCall
->        WHERE roll_no = r_no
->    ) THEN
->
->        INSERT INTO O_RollCall
->        VALUES(r_no, s_name);
->
->    END IF;
->
->    END LOOP;
->
->    CLOSE cur;
->
-> END //

```

Query OK, 0 rows affected (0.04 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL merge_rollcall();

Query OK, 0 rows affected (0.14 sec)

```
mysql> SELECT * FROM O_RollCall;
```

```
+-----+-----+
| roll_no | student_name |
+-----+-----+
| 1 | Rutuja |
| 2 | Amit |
| 3 | Sneha |
| 4 | Rahul |
| 5 | Pooja |
+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> -- STEP 3 : Create Library Table
```

```
mysql>
```

```
mysql> CREATE TABLE Library (
```

```
-> book_id INT PRIMARY KEY,
-> book_name VARCHAR(100),
-> author VARCHAR(100),
-> price INT
-> );
```

Query OK, 0 rows affected (0.27 sec)

```
mysql> -- STEP 4 : Create Library_Audit Table
```

```
mysql>
```

```
mysql> CREATE TABLE Library_Audit (
```

```
-> audit_id INT AUTO_INCREMENT PRIMARY KEY,
-> book_id INT,
```

```
-> old_book_name VARCHAR(100),  
-> old_author VARCHAR(100),  
-> old_price INT,  
-> operation_type VARCHAR(20),  
-> action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> INSERT INTO Library VALUES
```

```
-> (1,'DBMS','Korth',500),  
-> (2,'Java','James Gosling',700),  
-> (3,'Python','Guido',600);
```

Query OK, 3 rows affected (0.03 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE TRIGGER before_library_update
```

```
-> BEFORE UPDATE ON Library  
-> FOR EACH ROW  
-> BEGIN  
->  
-> INSERT INTO Library_Audit  
-> (book_id, old_book_name, old_author,  
-> old_price, operation_type)  
->  
-> VALUES  
-> (OLD.book_id, OLD.book_name,
```

-> OLD.author, OLD.price,

-> 'BEFORE UPDATE');

->

-> END //

Query OK, 0 rows affected (0.13 sec)

mysql>

mysql> DELIMITER ;

mysql> DELIMITER //

mysql>

mysql> CREATE TRIGGER after_library_update

-> AFTER UPDATE ON Library

-> FOR EACH ROW

-> BEGIN

->

-> INSERT INTO Library_Audit

-> (book_id, old_book_name, old_author,

-> old_price, operation_type)

->

-> VALUES

-> (OLD.book_id, OLD.book_name,

-> OLD.author, OLD.price,

-> 'AFTER UPDATE');

->

-> END //

Query OK, 0 rows affected (0.24 sec)

mysql>


```
mysql> DELIMITER ;
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_library_delete
    -> BEFORE DELETE ON Library
    -> FOR EACH ROW
    -> BEGIN
    ->
    -> INSERT INTO Library_Audit
    -> (book_id, old_book_name, old_author,
    -> old_price, operation_type)
    ->
    -> VALUES
    -> (OLD.book_id, OLD.book_name,
    -> OLD.author, OLD.price,
    -> 'BEFORE DELETE');
    ->
    -> END //
```

Query OK, 0 rows affected (0.19 sec)

```
mysql>
mysql> DELIMITER ;
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_library_delete
    -> AFTER DELETE ON Library
    -> FOR EACH ROW
    -> BEGIN
```

```

->
-> INSERT INTO Library_Audit
-> (book_id, old_book_name, old_author,
-> old_price, operation_type)
->
-> VALUES
-> (OLD.book_id, OLD.book_name,
-> OLD.author, OLD.price,
-> 'AFTER DELETE');
->
-> END //

```

Query OK, 0 rows affected (0.14 sec)

mysql>

mysql> DELIMITER ;

mysql> UPDATE Library

```
-> SET price = 800
```

```
-> WHERE book_id = 1;
```

Query OK, 1 row affected (0.04 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM Library

```
-> WHERE book_id = 2;
```

Query OK, 1 row affected (0.03 sec)

mysql> SELECT * FROM Library;

```
+-----+-----+-----+-----+
```

```
| book_id | book_name | author | price |
```

```
+-----+-----+-----+-----+
```

```
| 1 | DBMS | Korth | 800 |
```

```
| 3 | Python | Guido | 600 |
```

```
+-----+-----+-----+-----+
```

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM Library_Audit;
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
| audit_id | book_id | old_book_name | old_author | old_price | operation_type |  
action_time |
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

```
| 1 | 1 | DBMS | Korth | 500 | BEFORE UPDATE | 2026-05-07 00:41:44 |
```

```
| 2 | 1 | DBMS | Korth | 500 | AFTER UPDATE | 2026-05-07 00:41:44 |
```

```
| 3 | 2 | Java | James Gosling | 700 | BEFORE DELETE | 2026-05-07 00:41:54 |
```

```
|
```

```
| 4 | 2 | Java | James Gosling | 700 | AFTER DELETE | 2026-05-07 00:41:54 |
```

```
+-----+-----+-----+-----+-----+-----+-----+
```

4 rows in set (0.00 sec)

```
mysql> CREATE TABLE CUSTOMERS3 (
```

```
-> ID INT PRIMARY KEY,
```

```
-> NAME VARCHAR(50),
```

```
-> AGE INT,
```

```
-> ADDRESS VARCHAR(100),
```

```
-> SALARY DECIMAL(10,2)
```

```
-> );
```

Query OK, 0 rows affected (0.26 sec)

```
mysql> -- STEP 4 : Create Salary_Log Table
```

```
mysql>
```

```
mysql> CREATE TABLE Salary_Log (
```

```
-> log_id INT AUTO_INCREMENT PRIMARY KEY,  
-> customer_id INT,  
-> old_salary DECIMAL(10,2),  
-> new_salary DECIMAL(10,2),  
-> salary_difference DECIMAL(10,2),  
-> operation_type VARCHAR(20),  
-> log_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );
```

```
Query OK, 0 rows affected (0.27 sec)
```

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE TRIGGER trg_after_insert
```

```
-> AFTER INSERT ON CUSTOMERS3  
-> FOR EACH ROW  
-> BEGIN  
->  
-> INSERT INTO Salary_Log  
-> (customer_id, old_salary, new_salary,  
-> salary_difference, operation_type)  
->  
-> VALUES  
-> (NEW.ID, 0, NEW.SALARY,  
-> NEW.SALARY, 'INSERT');  
->  
-> END //
```

Query OK, 0 rows affected (0.04 sec)

mysql>

mysql> DELIMITER ;

mysql>

mysql>

mysql> DELIMITER //

mysql>

mysql> CREATE TRIGGER trg_after_update

-> AFTER UPDATE ON CUSTOMERS3

-> FOR EACH ROW

-> BEGIN

->

-> INSERT INTO Salary_Log

-> (customer_id, old_salary, new_salary,

-> salary_difference, operation_type)

->

-> VALUES

-> (NEW.ID, OLD.SALARY, NEW.SALARY,

-> NEW.SALARY - OLD.SALARY,

-> 'UPDATE');

->

-> END //

Query OK, 0 rows affected (0.02 sec)

mysql>

mysql> DELIMITER ;

mysql>

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE TRIGGER trg_after_delete
```

```
-> AFTER DELETE ON CUSTOMERS3
```

```
-> FOR EACH ROW
```

```
-> BEGIN
```

```
->
```

```
-> INSERT INTO Salary_Log
```

```
-> (customer_id, old_salary, new_salary,
```

```
-> salary_difference, operation_type)
```

```
->
```

```
-> VALUES
```

```
-> (OLD.ID, OLD.SALARY, 0,
```

```
-> 0 - OLD.SALARY,
```

```
-> 'DELETE');
```

```
->
```

```
-> END //
```

```
Query OK, 0 rows affected (0.14 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql> INSERT INTO CUSTOMERS3 VALUES
```

```
-> (1,'Rutuja',23,'Pune',40000),
```

```
-> (2,'Amit',25,'Mumbai',50000);
```

```
Query OK, 2 rows affected (0.01 sec)
```

```
Records: 2 Duplicates: 0 Warnings: 0
```

```
mysql> UPDATE CUSTOMERS3
```

-> SET SALARY = 60000

-> WHERE ID = 1;

Query OK, 1 row affected (0.12 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> DELETE FROM CUSTOMERS3

-> WHERE ID = 2;

Query OK, 1 row affected (0.03 sec)

mysql> SELECT * FROM CUSTOMERS3;

ID	NAME	AGE	ADDRESS	SALARY
1	Rutuja	23	Pune	60000.00

1 row in set (0.00 sec)

mysql> SELECT * FROM Salary_Log;

log_id	customer_id	old_salary	new_salary	salary_difference	operation_type	log_time	
1	1	0.00	40000.00	40000.00	INSERT	2026-05-07 00:45:50	
2	2	0.00	50000.00	50000.00	INSERT	2026-05-07 00:45:50	
3	1	40000.00	60000.00	20000.00	UPDATE	2026-05-07 00:46:17	
4	2	50000.00	0.00	-50000.00	DELETE	2026-05-07 00:46:43	

```
+-----+-----+-----+-----+-----+-----+-----+
--+
```

4 rows in set (0.00 sec)

mysql> -- STEP 3 : Create emp6 Table

mysql>

mysql> CREATE TABLE emp6 (

-> emp_no INT PRIMARY KEY,

-> salary DECIMAL(10,2)

->);

Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE increment_salary (

-> emp_no INT,

-> salary DECIMAL(10,2)

->);

Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO emp6 VALUES

-> (1,20000),

-> (2,30000),

-> (3,40000),

-> (4,50000),

-> (5,60000);

Query OK, 5 rows affected (0.13 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> DELIMITER //

mysql>

mysql> CREATE PROCEDURE increase_salary()

-> BEGIN

->

-> DECLARE done INT DEFAULT 0;

->

-> DECLARE e_no INT;

-> DECLARE sal DECIMAL(10,2);

->

-> DECLARE avg_sal DECIMAL(10,2);

->

-> -- Calculate Average Salary

-> SELECT AVG(salary)

-> INTO avg_sal

-> FROM emp6;

->

-> -- Cursor

-> DECLARE cur CURSOR FOR

-> SELECT emp_no, salary

-> FROM emp6

-> WHERE salary < avg_sal;

->

-> DECLARE CONTINUE HANDLER FOR NOT FOUND

-> SET done = 1;

->

-> OPEN cur;

->

-> read_loop: LOOP

```

->
->    FETCH cur INTO e_no, sal;
->
->    IF done = 1 THEN
->        LEAVE read_loop;
->    END IF;
->
->    -- Increase Salary by 10%
->    UPDATE emp6
->    SET salary = salary + (salary * 0.10)
->    WHERE emp_no = e_no;
->
->    -- Store Updated Salary
->    INSERT INTO increment_salary
->    SELECT emp_no, salary
->    FROM emp6
->    WHERE emp_no = e_no;
->
->    END LOOP;
->
->    CLOSE cur;
->
-> END //

```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'DECLARE cur CURSOR FOR

```

SELECT emp_no, salary

FROM emp6

WHERE salary ' at line 17

```

mysql>

mysql> DELIMITER ;

mysql> DELIMITER //

mysql>

mysql> CREATE PROCEDURE increase_salary()

-> BEGIN

->

-> DECLARE done INT DEFAULT 0;

->

-> DECLARE e_no INT;

-> DECLARE sal DECIMAL(10,2);

->

-> DECLARE avg_sal DECIMAL(10,2);

->

-> -- Cursor Declaration

-> DECLARE cur CURSOR FOR

-> SELECT emp_no, salary

-> FROM emp6

-> WHERE salary <

-> (SELECT AVG(salary) FROM emp6);

->

-> -- Handler

-> DECLARE CONTINUE HANDLER FOR NOT FOUND

-> SET done = 1;

->

-> -- Calculate Average Salary

-> SELECT AVG(salary)

-> INTO avg_sal

```
-> FROM emp6;
->
-> OPEN cur;
->
-> read_loop: LOOP
->
->     FETCH cur INTO e_no, sal;
->
->     IF done = 1 THEN
->         LEAVE read_loop;
->     END IF;
->
->     -- Increase salary by 10%
->     UPDATE emp6
->     SET salary = salary + (salary * 0.10)
->     WHERE emp_no = e_no;
->
->     -- Insert updated salary into increment_salary table
->     INSERT INTO increment_salary
->     SELECT emp_no, salary
->     FROM emp6
->     WHERE emp_no = e_no;
->
-> END LOOP;
->
-> CLOSE cur;
->
-> END //
```

Query OK, 0 rows affected (0.11 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL increase_salary();

Query OK, 0 rows affected (0.36 sec)

mysql> SELECT * FROM emp6;

+-----+-----+

| emp_no | salary |

+-----+-----+

| 1 | 22000.00 |

| 2 | 33000.00 |

| 3 | 40000.00 |

| 4 | 50000.00 |

| 5 | 60000.00 |

+-----+-----+

5 rows in set (0.00 sec)

mysql> SELECT * FROM increment_salary;

+-----+-----+

| emp_no | salary |

+-----+-----+

| 1 | 22000.00 |

| 2 | 33000.00 |

+-----+-----+

2 rows in set (0.00 sec)

mysql> -- STEP 3 : Create studentt8 Table

mysql>

mysql> CREATE TABLE studentt8 (

-> Roll INT PRIMARY KEY,

-> Name VARCHAR(50),

-> Attendance DECIMAL(5,2),

-> Status VARCHAR(20)

->);

Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO studentt8 VALUES

-> (1,'Rutuja',80,"),

-> (2,'Amit',60,"),

-> (3,'Sneha',90,"),

-> (4,'Rahul',70,");

Query OK, 4 rows affected (0.23 sec)

Records: 4 Duplicates: 0 Warnings: 0

mysql> DELIMITER //

mysql>

mysql> CREATE PROCEDURE check_attendance(

-> IN r_no INT

->)

->

-> BEGIN

->

-> DECLARE att DECIMAL(5,2);

->

```

-> -- Exception Handling
-> DECLARE EXIT HANDLER FOR SQLEXCEPTION
-> BEGIN
->     SELECT 'Error Occurred' AS Message;
-> END;
->
-> -- Check Roll Number Exists
-> IF EXISTS (
->     SELECT *
->     FROM studentt8
->     WHERE Roll = r_no
-> ) THEN
->
->     -- Get Attendance
->     SELECT Attendance
->     INTO att
->     FROM studentt8
->     WHERE Roll = r_no;
->
->     -- Check Attendance
->     IF att < 75 THEN
->
->         UPDATE studentt8
->         SET Status = 'Detained'
->         WHERE Roll = r_no;
->
->         SELECT 'Term not granted' AS Message;
->

```

```

-> ELSE
->
-> UPDATE studentt8
-> SET Status = 'Not Detained'
-> WHERE Roll = r_no;
->
-> SELECT 'Term granted' AS Message;
->
-> END IF;
->
-> ELSE
->
-> SELECT 'Roll Number Not Found' AS Message;
->
-> END IF;
->
-> END //

```

Query OK, 0 rows affected (0.03 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL check_attendance(2);

```
+-----+
```

```
| Message      |
```

```
+-----+
```

```
| Term not granted |
```

```
+-----+
```

1 row in set (0.03 sec)

Query OK, 0 rows affected (0.04 sec)

```
mysql> SELECT * FROM studentt8;
```

```
+-----+-----+-----+-----+
| Roll | Name  | Attendance | Status |
+-----+-----+-----+-----+
| 1 | Rutuja | 80.00 |   |
| 2 | Amit  | 60.00 | Detained |
| 3 | Sneha | 90.00 |   |
| 4 | Rahul | 70.00 |   |
+-----+-----+-----+-----+
```

4 rows in set (0.00 sec)

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE FUNCTION Age_calc(dob DATE)
```

```
-> RETURNS INT
```

```
-> DETERMINISTIC
```

```
->
```

```
-> BEGIN
```

```
->
```

```
-> DECLARE years INT;
```

```
->
```

```
-> -- Calculate age in years
```

```
-> SET years = TIMESTAMPDIFF(YEAR, dob, CURDATE());
```

```
->
```

```
-> RETURN years;
```

->

-> END //

Query OK, 0 rows affected (0.12 sec)

mysql>

mysql> DELIMITER ;

mysql> DELIMITER //

mysql>

mysql> CREATE PROCEDURE Age_details(

-> IN dob DATE,

-> OUT months INT,

-> OUT days INT

->)

->

-> BEGIN

->

-> DECLARE total_months INT;

->

-> -- Calculate total months

-> SET total_months =

-> TIMESTAMPDIFF(MONTH, dob, CURDATE());

->

-> -- Remaining months after years

-> SET months = total_months % 12;

->

-> -- Remaining days

-> SET days =

-> DATEDIFF(

```

->    CURDATE(),
->    DATE_ADD(
->        dob,
->        INTERVAL total_months MONTH
->    )
-> );
->
-> END //

```

Query OK, 0 rows affected (0.03 sec)

mysql>

mysql> DELIMITER ;

mysql> SELECT Age_calc('2002-05-10') AS Age_Years;

```

+-----+
| Age_Years |
+-----+
|    23    |
+-----+

```

1 row in set (0.02 sec)

mysql> CALL Age_details(

```

->    '2002-05-10',
->    @months,
->    @days
-> );

```

Query OK, 0 rows affected (0.02 sec)

mysql> SELECT

-> @months AS Months,

-> @days AS Days;

+-----+-----+

| Months | Days |

+-----+-----+

| 11 | 27 |

+-----+-----+

1 row in set (0.00 sec)

mysql> -- STEP 3 : Create Stud_Marks Table

mysql>

mysql> CREATE TABLE Stud_Marks (

-> Roll INT PRIMARY KEY,

-> Name VARCHAR(50),

-> Total_Marks INT

->);

Query OK, 0 rows affected (0.27 sec)

mysql> -- STEP 4 : Create Result Table

mysql>

mysql> CREATE TABLE Result (

-> Roll INT,

-> Name VARCHAR(50),

-> Class VARCHAR(50)

->);

Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Stud_Marks VALUES

-> (1,'Rutuja',1200),

-> (2,'Amit',950),

-> (3,'Sneha',850),

-> (4,'Rahul',700);

Query OK, 4 rows affected (0.01 sec)

Records: 4 Duplicates: 0 Warnings: 0

mysql> DELIMITER //

mysql>

mysql> CREATE PROCEDURE proc_Grade()

-> BEGIN

->

-> DECLARE done INT DEFAULT 0;

->

-> DECLARE r_no INT;

-> DECLARE s_name VARCHAR(50);

-> DECLARE marks INT;

-> DECLARE class_name VARCHAR(50);

->

-> DECLARE cur CURSOR FOR

-> SELECT Roll, Name, Total_Marks

-> FROM Stud_Marks;

->

-> DECLARE CONTINUE HANDLER FOR NOT FOUND

-> SET done = 1;

->

-> OPEN cur;

->

```
-> read_loop: LOOP
->
->   FETCH cur INTO r_no, s_name, marks;
->
->   IF done = 1 THEN
->     LEAVE read_loop;
->   END IF;
->
->   -- Categorization
->   IF marks >= 990 AND marks <= 1500 THEN
->     SET class_name = 'Distinction';
->
->   ELSEIF marks >= 900 AND marks <= 989 THEN
->     SET class_name = 'First Class';
->
->   ELSEIF marks >= 825 AND marks <= 899 THEN
->     SET class_name = 'Higher Second Class';
->
->   ELSE
->     SET class_name = 'Fail';
->
->   END IF;
->
->   -- Insert into Result Table
->   INSERT INTO Result
->     VALUES(r_no, s_name, class_name);
->
-> END LOOP;
```

->

-> CLOSE cur;

->

-> END //

Query OK, 0 rows affected (0.03 sec)

mysql>

mysql> DELIMITER ;

mysql> CALL proc_Grade();

Query OK, 0 rows affected (0.06 sec)

mysql> SELECT * FROM Stud_Marks;

Roll	Name	Total_Marks
1	Rutuja	1200
2	Amit	950
3	Sneha	850
4	Rahul	700

4 rows in set (0.00 sec)

mysql> SELECT * FROM Result;

Roll	Name	Class
1	Rutuja	Distinction
2	Amit	First Class

| 3 | Sneha | Higher Second Class |

| 4 | Rahul | Fail |

+-----+-----+-----+-----+

4 rows in set (0.00 sec)

mysql> -- STEP 3 : Create Borrower Table

mysql>

mysql> CREATE TABLE Borrower (

-> Rollin INT,

-> Name VARCHAR(50),

-> DateofIssue DATE,

-> NameofBook VARCHAR(100),

-> Status CHAR(1)

->);

Query OK, 0 rows affected (0.05 sec)

mysql> -- STEP 4 : Create Fine Table

mysql>

mysql> CREATE TABLE Fine (

-> Roll_no INT,

-> Date DATE,

-> Amt DECIMAL(10,2)

->);

Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Borrower VALUES

-> (1,'Rutuja','2026-04-01','DBMS','I'),

-> (2,'Amit','2026-03-01','Java','I');

Query OK, 2 rows affected (0.03 sec)

Records: 2 Duplicates: 0 Warnings: 0

```
mysql> DELIMITER //
```

```
mysql>
```

```
mysql> CREATE PROCEDURE calculate_fine(
```

```
    -> IN r_no INT,
```

```
    -> IN b_name VARCHAR(100)
```

```
    -> )
```

```
    ->
```

```
    -> BEGIN
```

```
    ->
```

```
    -> DECLARE issue_date DATE;
```

```
    -> DECLARE days_late INT;
```

```
    -> DECLARE fine_amt DECIMAL(10,2);
```

```
    ->
```

```
    -> -- Get Date of Issue
```

```
    -> SELECT DateofIssue
```

```
    -> INTO issue_date
```

```
    -> FROM Borrower
```

```
    -> WHERE Rollin = r_no
```

```
    -> AND NameofBook = b_name;
```

```
    ->
```

```
    -> -- Calculate number of days
```

```
    -> SET days_late =
```

```
    -> DATEDIFF(CURDATE(), issue_date);
```

```
    ->
```

```
    -> -- Fine Calculation
```

```

-> IF days_late BETWEEN 15 AND 30 THEN
->
->     SET fine_amt = days_late * 5;
->
-> ELSEIF days_late > 30 THEN
->
->     SET fine_amt =
->     (30 * 5) +
->     ((days_late - 30) * 50);
->
-> ELSE
->
->     SET fine_amt = 0;
->
-> END IF;
->
-> -- Update Book Status
-> UPDATE Borrower
-> SET Status = 'R'
-> WHERE Rollin = r_no
-> AND NameofBook = b_name;
->
-> -- Insert Fine Details if Fine Exists
-> IF fine_amt > 0 THEN
->
->     INSERT INTO Fine
->     VALUES(
->         r_no,

```

```

-> CURDATE(),
-> fine_amt
-> );
->
-> END IF;
->
-> END //

```

Query OK, 0 rows affected (0.03 sec)

```

mysql>
mysql> DELIMITER ;
mysql> CALL calculate_fine(1,'DBMS');
Query OK, 1 row affected (0.04 sec)

```

```

mysql> SELECT * FROM Borrower;
+-----+-----+-----+-----+-----+
| Rollin | Name | DateofIssue | NameofBook | Status |
+-----+-----+-----+-----+-----+
| 1 | Rutuja | 2026-04-01 | DBMS | R |
| 2 | Amit | 2026-03-01 | Java | I |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

```

```

mysql> SELECT * FROM Fine;
+-----+-----+-----+
| Roll_no | Date | Amt |
+-----+-----+-----+
| 1 | 2026-05-07 | 450.00 |

```

```
+-----+-----+-----+
```

1 row in set (0.00 sec)

mysql>